Ecole Supérieure d'Informatique et du Numérique
COLLEGE OF ENGINEERING & ARCHITECTURE

Introduction to Cybersecurity

NETWORK SECURITY

PRESENTED BY:
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Lecture: Introduction To CyberSecurity

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Coordinator of the Cybersecurity Track:

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7	Application	Human-computer interaction | HTTP, DNS, FTP, POP3
6	Presentation	Data representation and encryption | SSH, SSL, IMAP, FTP, JPEG
5	Session	Connection maintenance | API's, sockets
4	Transport	End-to-end connections | TCP, UDP
3	Network	Logical addressing for data path using packets | IP, ICMP, IGMP
2	Data	Physical addressing, formatting data in frames | switch, bridge, ethernet
1	Physical	Physical transmission of signals, media, or binary raw bits | coax, fiber, wireless

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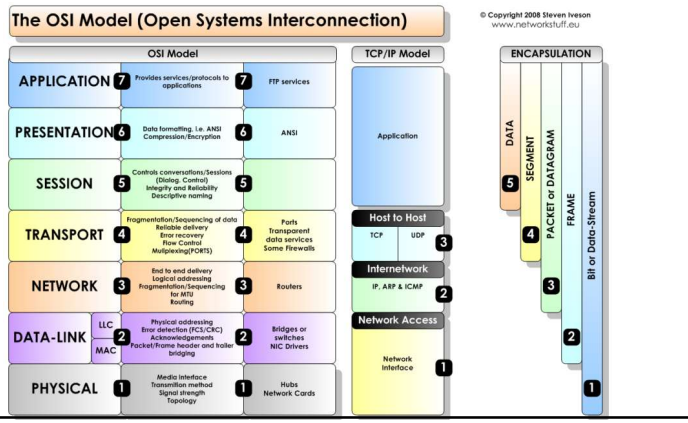
OSI vs TCP/IP Model



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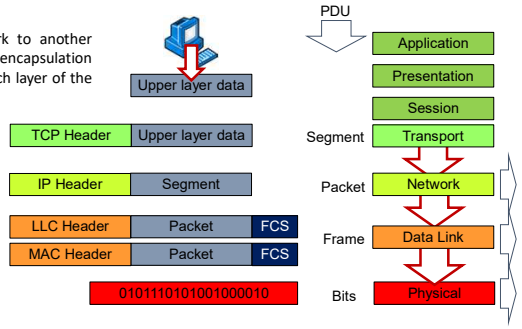
Network Security

Encapsulation des données

When a host transmits data across a network to another device, the data goes through a process called encapsulation and is wrapped with protocol information at each layer of the OSI model.

The Figure shows the Protocol Data Unit (PDUs) and how they attach control information to each layer.



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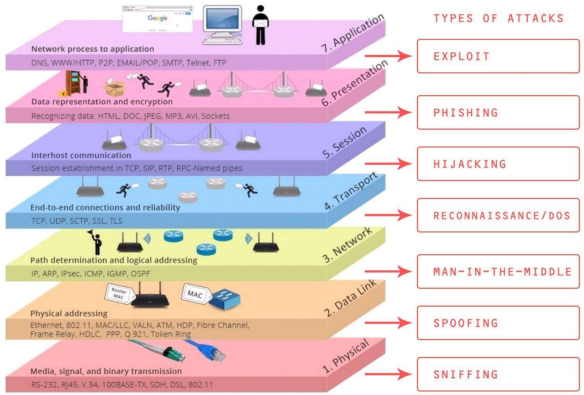
Example : Network OSI Models Attacks



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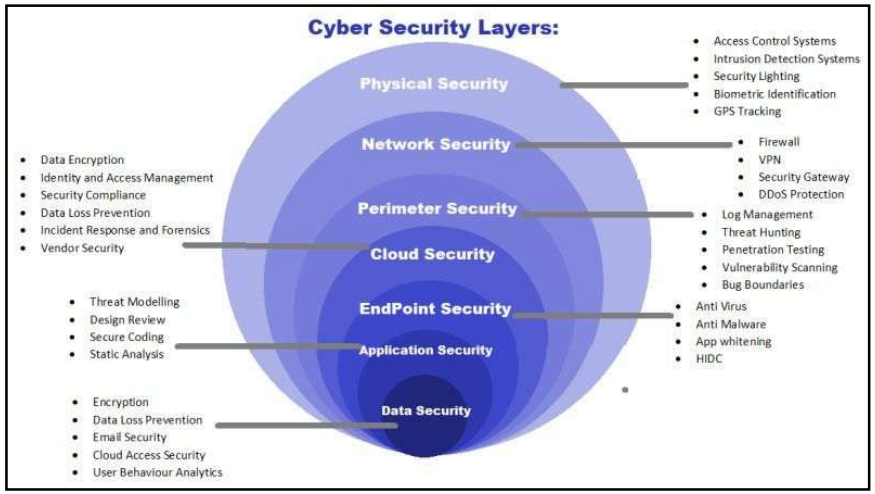

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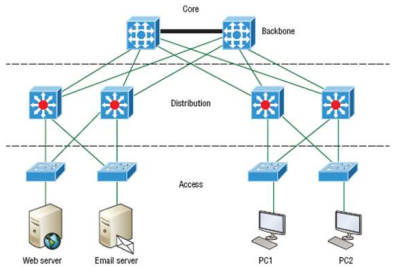
Three-layer hierarchical model

The hierarchical model can help you design, implement, and maintain a scalable, reliable, and cost-effective hierarchical internetwork. The model defines three hierarchy layers, as shown in the Figure, each with specific functions.



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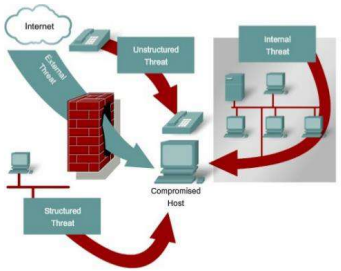
Network Security

Network Attack Vectors

- **External Attacks**
- **Insider Attacks**
- **Structured Attacks**
- **Unstructured attacks**



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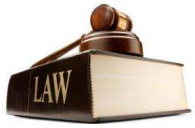
The laws of CyberSecurity

Legislation:
Several laws are established by the federal and government to protect users from the threats of cybercrime:
1- The healt Insurance Portability and Accountability 1996
2- The sarbanes-Oxley Act of 2002
3- The Gramm-Leach-Blilely Act
4- US PATRIOT ACT 2001

In Morocco we have:
1- Law 53-05 on the electronic exchange of legal data
2- Law 09-08 on the automated processing of personal data



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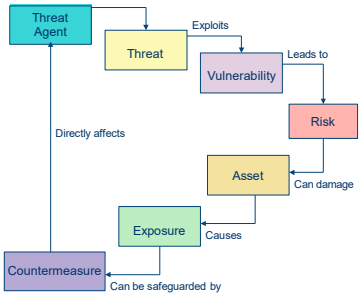
Common network security terms:



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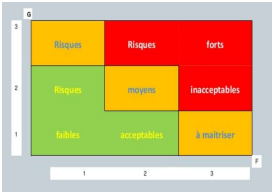
CyberSecurity Risk

Risk Management
Definition:
The assessment and quantification process and the establishment of an acceptable level of risk to the organization

Risk = Probability * Impact

Stages of risk management:
- **Risk analysis**
- **Assessed and identified threats**
- **Assessed and identified vulnerabilities**
- **Establish a Security Policy (countermeasure)**



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CyberSecurity Risk

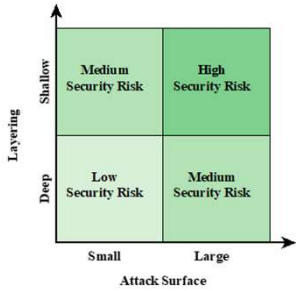

Figure 1.4 Defense in Depth and Attack Surface

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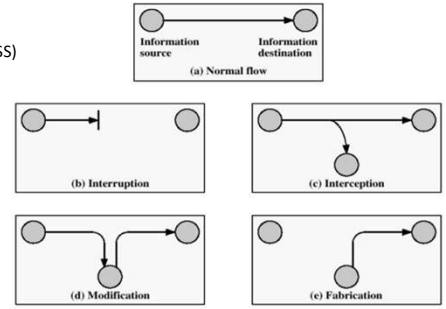
Attackers' tools

- Port/address scan (nmap, Nessus)
- Vulnerability Scanner (MetaSploit, Core Impact, ISS)
- Sniffers (Wire shark, tcpdump, Snort)
- Cracking tools (Cain, wepCrack, jhon the ripper)
- Malware (viruses, worms, trojon)
- Hijackin tools (Netcat, MetaSploit)



Aims of attacks

- **Interruption**: aims at the availability of information
- **Interception**: aims at the confidentiality of information
- **Modification**: aims at the integrity of information
- **Manufacturing**: aims at the authenticity of the information

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Some known attacks:

- **ARP attack**
- **Brute Force Attack**
- **Malware**
- **Flooding**
- **Sniffers**
- **Spoofing**
- **Redirected Attacks**
- **Tunneling Attack**
- **Owasp Attack**

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Example of real attacks

WannaCry – Considered the largest ransomware cyberattack in history, WannaCry infected more than 300,000 computers in more than 150 countries within hours.
Among its victims: FedEx, Renault, or the Russian Ministry of the Interior.

Deloitte – September – For nearly 6 months, the prestigious consulting and audit firm was the victim of a major cyberattack during which hackers managed to access private information, such as emails exchanged between the firm and its clients. The hackers used the username and password of an administrator account, allowing them to access Miscrosoft's Azure Cloud, the platform hosting part of Deloitte's data.

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Example of real attacks

Netflix – *September* – The streaming platform Netflix has been the victim of a major hack, more precisely a scam campaign directly targeting its users millions of them have received emails from the supportnetflix@com address, inviting them to communicate their bank details to prevent their account from being closed. As usual, everything had been carefully thought out to deceive the victims: website with the graphic charter of the real platform, use of a tone and design similar to those used by Netflix.

Uber – *November* – About a year ago, nearly 57 million user accounts on the Uber platform were hacked. The American company, world leader in ride-hailing, would then have made the decision to pay a ransom to hackers of $ 100,000 in exchange for the destruction of the hacked data, without having the assurance that it is actually carried out.

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Introduction

2021 : Marriott

- For the second time in two years, the popular hotel chain suffered a data breach.
- On March 31, Marriott issued a statement revealing that the information of 5.2 million guests was accessed using the login credentials of two employees at a franchised property.
- The compromised information may have involved contact and customer loyalty account information, but not passwords.

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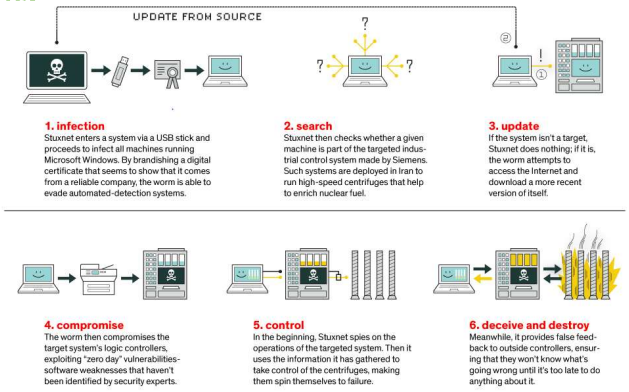
Exemple d'attaques réel

Stuxnet : A splendid com
1.Complex multi-faceted r
2.Targeted uranium enrich
for years.
3.Initiated debate on digit
4.Rapid worldwide prolife
computers.
5.Designed to execute a sp
6.Targeted computers ope
7.Once installed, reprogra
8.Caused physical damage
uranium enrichment cent



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Who to protect yourself from

An attack is a threat made by people:

- **Hackers:** who take advantage of accessing other people's information systems. Some simply leave their "mark". Others called "robbers" crash systems, steal or damage data, or interrupt business.
- **Unsuspecting personnel:** unintentionally being the source of security breaches (virus contamination, software installation, etc.)
- **Disgruntled users:** abuse their rights and cause damage to certain services.

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CyberSca Terminologies

Attacks = Motive (Goal) + Method + Vulnerability

- A motive originates out of the notion that the target system stores or processes something valuable, and this leads to the threat of an attack on the system
- Attackers try various tools and attack techniques to exploit vulnerabilities in a computer system or its security policy and controls in order to fulfil their motives

Motives behind information security attacks

- Disrupting business continuity
- Stealing information and manipulating data
- Creating fear and chaos by disrupting critical infrastructures
- Causing financial loss to the target
- Propagating religious or political beliefs
- Achieving a state's military objectives
- Damaging the reputation of the target
- Taking revenge
- Demanding ransom

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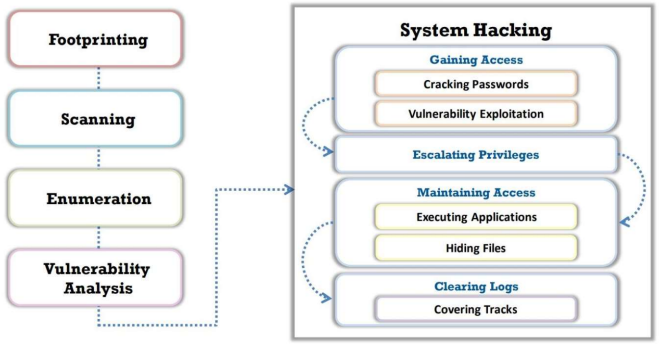
Anatomy the attack



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Tactics, Techniques, and Procedures TTPs

The term Tactics, Techniques, and Procedures (TTPs) refers to the patterns of activities and methods associated with specific threat actors or groups of threat actors

Tactics

- "Tactics" are the guidelines that describe the way an attacker performs the attack from beginning to the end
- This guideline consists of the various tactics for information gathering to perform initial exploitation, privilege escalation, and lateral movement, and to deploy measures for persistent access to the system and other purposes

Techniques

- "Techniques" are the technical methods used by an attacker to achieve intermediate results during the attack
- These techniques include initial exploitation, setting up and maintaining command and control channels, accessing the target infrastructure, covering the tracks of data exfiltration, and others

Procedures

- "Procedures" are organizational approaches that threat actors follow to launch an attack
- The number of actions usually differs depending on the objectives of the procedure and threat actor group

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Attack Method

- **Search for the machine by:** ping, ...
- **Determine the intermediate nodes:** traceroute, ..
- **Active Services Search:** TCP SCAN, SYN SCAN, UDP SCAN
- **Identification of machines and software, ID field, Stack finger print, obtaining software version and configuration, obtaining information through accessible services.**
- **Attempts to exploit vulnerabilities:** buffer overflow, attacks on web services, horses of three, etc.
- **Target Attack**

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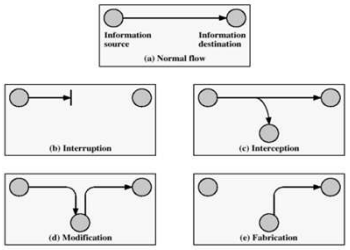
Attackers' tools

- Scanne de port/address (nmap, Nessus)
- Scanne de vulnérabilité (MetaSploit, Core Impact, ISS)
- Sniffers (Wire shark, tcpdump, Snort)
- Outils de cracking (Cain, wepCrack, jhon the ripper)
- Malware (virus, vers, trojon)
- Outils de hijackin (Netcat, MetaSploit)

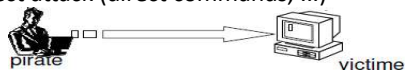


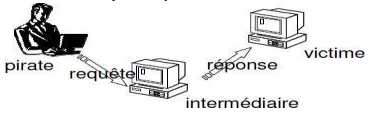
Goals of the attacks

- **Interruption: refers to the availability of information**
- **Interception: aims at the confidentiality of information**
- **Modification: aims at the integrity of the information**
- **Manufacturing: aims for the authenticity of the information**

Types of Attacks

- Direct attack (direct commands, ...)



- Bounce Indirect Attack (hides the identity of the attacker)

- Indirect Attack by Response